COMPUTER VISION · LECTURE 3

Homographies and RANSAC

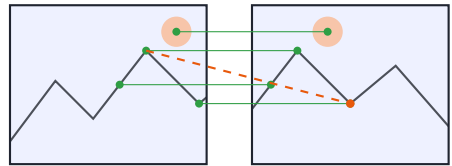
Aligning photos into a panorama, even when some matches are wrong

A demo deck for the Co-learner README

From two photos to one panorama

1. **Match** features that appear in both photos.
2. **Estimate** the transform that carries one photo onto the other.
3. **Warp** the photo into the shared frame and **blend** the overlap.

This lecture: step 2, and how to survive the wrong matches from step 1.



A homography maps a plane to a plane

In homogeneous coordinates a point $\mathbf{x} = (x, y, 1)^\top$ in one image maps to

$$\mathbf{x}' \sim H\mathbf{x}, \qquad H = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}$$

where $\sim$ means *equal up to scale*: H and cH describe the same mapping.

- 9 entries, 1 lost to scale: **8 degrees of freedom**.
- Exact when the scene is a plane, or when the camera only rotates about its center.
- Back to pixels: $x' = \dfrac{h_{11}x + h_{12}y + h_{13}}{h_{31}x + h_{32}y + h_{33}}$, $y' = \dfrac{h_{21}x + h_{22}y + h_{23}}{h_{31}x + h_{32}y + h_{33}}$.

Each match gives two linear equations

Multiply out the division. With $\mathbf{h} = (h_{11}, h_{12}, h_{13}, h_{21}, h_{22}, h_{23}, h_{31}, h_{32}, h_{33})^\top$, one match $(x, y) \leftrightarrow (x', y')$ gives

$$\begin{bmatrix} -x & -y & -1 & 0 & 0 & 0 & xx' & yx' & x' \\ 0 & 0 & 0 & -x & -y & -1 & xy' & yy' & y' \end{bmatrix} \mathbf{h} = \mathbf{0}.$$

- Linear in $\mathbf{h}$, even though the mapping itself is not.
- 8 unknowns (up to scale) and 2 equations per match: at least **4 matches**, no 3 of them on a line.

The Direct Linear Transform (DLT)

Stack the rows of all $n \geq 4$ matches into a $2n \times 9$ matrix A and solve

$$\min_{\mathbf{h}} \|A\mathbf{h}\| \quad \text{subject to} \quad \|\mathbf{h}\| = 1.$$

With the SVD $A = U\Sigma V^\top$, the answer is the **last column of** V: the right singular vector with the smallest singular value.

The constraint $\|\mathbf{h}\| = 1$ rules out the useless answer $\mathbf{h} = \mathbf{0}$.

Recipe

1. build A from the matches
2. $U, \Sigma, V^\top = \mathrm{svd}(A)$
3. $\mathbf{h} \leftarrow$ last row of $V^\top$
4. reshape $\mathbf{h}$ to 3×3

Normalize the points before you solve

Pixel coordinates near 1000 make the entries of A range from 1 to $xx' \approx 10^6$, so the SVD is badly conditioned.

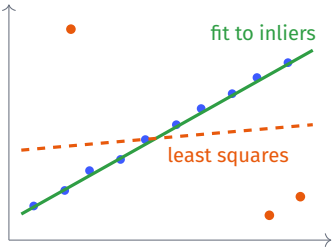
Normalized DLT

1. Find T (and T') that move the points' centroid to the origin and scale them so their mean distance from it is $\sqrt{2}$.
2. Run the DLT on $\hat{\mathbf{x}} = T\mathbf{x}$ and $\hat{\mathbf{x}}' = T'\mathbf{x}'$ to get $\hat{H}$.
3. Undo the normalization: $H = T'^{-1}\,\hat{H}\,T$.

Same answer for perfect data; far more stable with noise.

One wrong match ruins least squares

- Least squares weighs every match, so a single far-off point pulls the whole fit.
- Feature matching always produces some wrong matches (*outliers*).
- We need a fit that **ignores** points it cannot explain.



RANSAC: random sample consensus

Repeat N times

1. Pick **4 matches** at random.
2. Fit H to them with the DLT.
3. Count the **inliers**: matches with $\|\mathbf{x}'_i - H(\mathbf{x}_i)\| < \varepsilon$ pixels.

Keep the H with the most inliers, then **refit** it on all of its inliers.

- $H(\mathbf{x})$: apply H, then divide by the third coordinate.
- ε is a few pixels (for example 2 to 5).
- A sample of 4 inliers predicts the other inliers; a sample with an outlier usually predicts almost nothing.

How many samples are enough?

If a fraction w of the matches are inliers, a sample of s matches is all inliers with probability w^s. To succeed at least once with probability p:

$$N = \left\lceil \frac{\log(1-p)}{\log\left(1-w^s\right)} \right\rceil$$

inlier fraction w	s	N $(p = 0.99)$
0.9	4	5
0.7	4	17
0.5	4	72
0.3	4	567

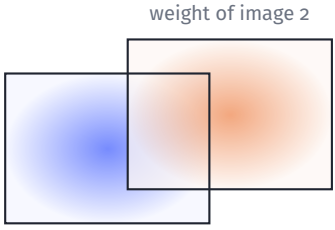
- N depends on w and s, not on how many matches there are.
- Fewer points per sample help: that is why we use the minimal 4.

Warp, then blend

Backward warping. For every pixel $\mathbf{p}$ of the output canvas, look up the source image at $H^{-1}\mathbf{p}$ and interpolate (bilinear). Every output pixel gets a value; no holes.

Feathering. Weight each image by its pixel's distance to that image's border (a distance transform), then average:

$$I(\mathbf{p}) = \frac{\sum_k w_k(\mathbf{p})\, I_k(\mathbf{p})}{\sum_k w_k(\mathbf{p})}$$



high in each image's middle, zero at its border: the overlap cross-fades, so no seam

Summary

- A homography H (8 degrees of freedom) maps one view of a plane, or of a rotating camera, onto another.
- The DLT turns 4 or more matches into $A\mathbf{h} = \mathbf{0}$; the SVD gives $\mathbf{h}$. Normalize first.
- RANSAC fits H to random minimal samples and keeps the one most matches agree with.
- Backward warping fills the canvas; distance-weighted blending hides the seams.

Next lecture: stereo and depth.